

CHAPTER 4: CAPACITOR

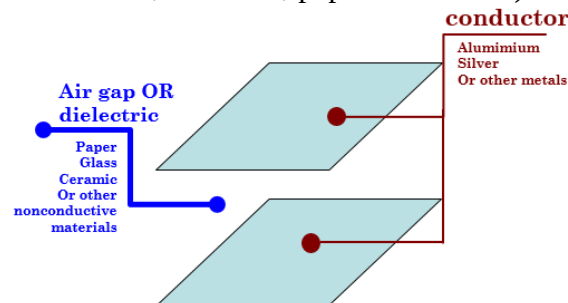
At the end of this chapter, you'll be able to:

SUBTOPIC	LEARNING OUTCOMES
4.1	Capacitance and capacitors in series and parallel a) Define and use capacitance, $C = \frac{Q}{V}$. b) Use the effective capacitance of various arrangements for: Series: $\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots \frac{1}{C_n}$ Parallel: $C_{eff} = C_1 + C_2 + C_3 + \cdots C_n$ <i>*limited to four capacitors.</i> <i>*included calculation of charge and voltage in the circuit.</i>
4.2	Charging and discharging of capacitors a) State physical meaning of time constant, τ and use $\tau = RC$. b) Sketch and explain the characteristics of $Q - t$ and $I - t$ graph for charging and discharging of a capacitor. c) Use: For discharging: $Q = Q_0 e^{-\frac{t}{RC}}$ For charging: $Q = Q_0 \left[1 - e^{-\frac{t}{RC}} \right]$ d) Determine the time constant of an RC circuit.

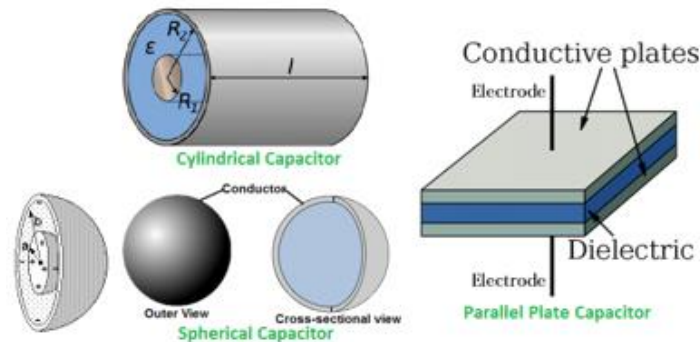
4.1 Capacitance And Capacitors In Series And Parallel

Capacitor

- A **device** that is capable of **storing electric charges** or electric potential energy.
- It **consists of two conducting plates separated** by a small air gap or a thin insulator (called a **dielectric** such as mica, ceramics, paper or even oil).



- The conducting plates could be in the shape of, cylindrical, spherical, parallel plate.



- The electrical symbol for a capacitor is:



Capacitance

- Capacitance is defined as **the ratio of the charge on either plate to the potential difference** between them.

$$C = \frac{Q}{V}$$

where

C : Capacitance

Q : Magnitude of charge on one of the plates

V : Potential difference between the plates

- It is a measure of the **ability of a capacitor to store charge**.
- The **S.I. unit** is **Farad (F)**.
- 1 Farad is defined as **the charge of 1 coulomb stored on each of the conducting plates as a result of a potential difference of 1 volt between the two plates**.
- The capacitance for a capacitor does not change unless it is designed to be a variable capacitor.
- The **charges stored (Q)** is **directly proportional to the potential difference (V)** across the conducting plate: $Q \propto V$
- A capacitor with a large capacitance can hold more charges than one with a smaller capacitance for the same potential difference applied across them.

Example 1

The charges on the plates of capacitor are $+120 \mu\text{C}$ and $-120 \mu\text{C}$ when they are connected to a potential difference of 6.0 V . Calculate

- The capacitance of the capacitor.
- The potential difference across the capacitor if the charge on each plate is $64 \mu\text{C}$.

Answers:

a) $Q = 120 \mu\text{C}; V = 6.0 \text{ V}$

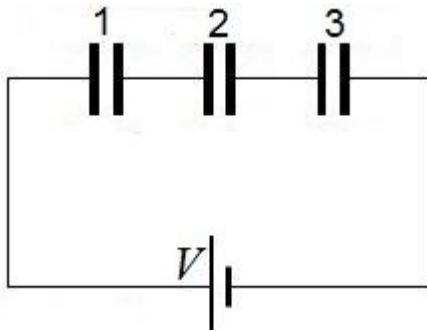
$$C = \frac{Q}{V} = \frac{120 \mu}{6.0}$$

$$C = 20 \mu\text{F}$$

b) $C = 20 \mu\text{F}; Q = 64 \mu\text{C}$

$$V = \frac{Q}{C} = \frac{64 \mu}{20 \mu}$$

$$V = 3.2 \text{ V}$$

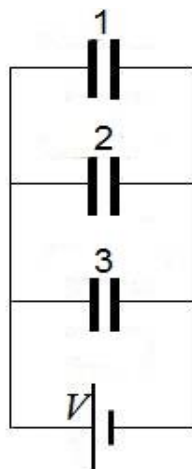
Capacitors in Series and Parallel**Capacitors in series**

$$Q = Q_1 = Q_2 = Q_3$$

$$V = V_1 + V_2 + V_3$$

$$\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

- Regardless of their capacitance, all capacitors in series contain charges of same magnitude.
- Total voltage, V is shared by the capacitors.
- Value of C_{eff} is always smaller than the smallest capacitance in the combination.

Capacitors in Parallel

$$Q = Q_1 + Q_2 + Q_3$$

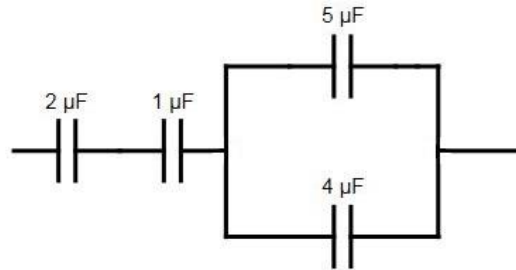
$$V = V_1 = V_2 = V_3$$

$$C_{eff} = C_1 + C_2 + C_3$$

- They have the same voltage, V across their plates.
- Charge, Q_1 and Q_2 on each capacitors are different.
- Value of C_{eff} is always larger than the largest individual capacitance.in the combination.

Example 2

Find the equivalent capacitance of the arrangement of capacitors shown in figure.



Answers:

Let $C_1 = 2 \mu F$; $C_2 = 1 \mu F$; $C_3 = 5 \mu F$ and $C_4 = 4 \mu F$

C_3 and C_4 are in parallel

$$C_{34} = C_3 + C_4$$

$$C_{34} = 5\mu + 4\mu$$

$$C_{34} = 9 \mu F$$

C_1 , C_2 and C_{34} are in series

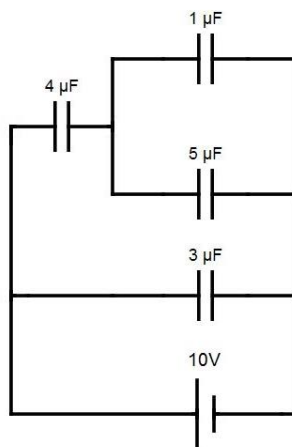
$$\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_{34}}$$

$$\frac{1}{C_{eff}} = \frac{1}{2\mu} + \frac{1}{1\mu} + \frac{1}{9\mu}$$

$$C_{eff} = 0.62 \mu F$$

Example 3

Find the equivalent capacitance of the arrangement of capacitors shown in figure.



Answers:

Let $C_1 = 4 \mu F$; $C_2 = 1 \mu F$; $C_3 = 5 \mu F$ and $C_4 = 3 \mu F$

C_2 and C_3 are in parallel

$$C_{23} = C_2 + C_3$$

$$C_{23} = 1\mu + 5\mu$$

$$C_{23} = 6 \mu F$$

C_1 and C_{23} are in series

$$\frac{1}{C_{123}} = \frac{1}{C_1} + \frac{1}{C_{23}}$$

$$\frac{1}{C_{123}} = \frac{1}{4\mu} + \frac{1}{6\mu}$$

$$C_{123} = 2.4 \mu F$$

C_{123} and C_4 are in parallel

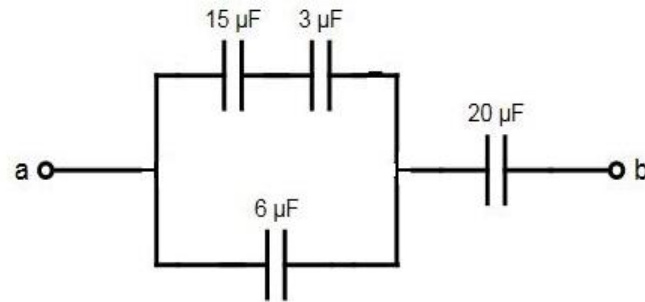
$$C_{eff} = C_{123} + C_4$$

$$C_{eff} = 2.4\mu + 3\mu$$

$$C_{eff} = 5.4 \mu F$$

Example 4

Four capacitors are connected as shown in figure below. Calculate:



- The equivalent capacitance between point a and b.
- The charge on each capacitor if $V_{ab} = 15V$.
- The voltage across each capacitor.

Answers:

- a) Let $C_1 = 15 \mu F$; $C_2 = 3 \mu F$; $C_3 = 6 \mu F$ and $C_4 = 20 \mu F$

C_1 and C_2 are in series

$$\frac{1}{C_{12}} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\frac{1}{C_{12}} = \frac{1}{15\mu} + \frac{1}{3\mu}$$

$$C_{12} = 2.5 \mu F$$

C_{12} and C_3 are in parallel

$$C_{123} = C_{12} + C_3$$

$$C_{123} = 2.5\mu + 6\mu$$

$$C_{123} = 8.5 \mu F$$

C_{123} and C_4 are in series

$$\frac{1}{C_{eff}} = \frac{1}{C_{123}} + \frac{1}{C_4}$$

$$\frac{1}{C_{eff}} = \frac{1}{8.5\mu} + \frac{1}{20\mu}$$

$$C_{eff} = 5.96 \mu F$$

- b) $Q_T = C_{eff} V_{ab} = (5.96 \mu)(15) = 89.4 \mu C$

C_{123} and C_4 are in series so $Q_T = Q_{123} = Q_4 = 89.4 \mu C$

$$V_4 = \frac{Q_4}{C_4} = \frac{89.4 \mu}{20 \mu} = 4.47 V$$

$$V_{ab} = V_{123} + V_4$$

$$15 = V_{123} + 4.45$$

$$V_{123} = 10.53 V$$

C_{12} and C_3 are in parallel so $V_{123} = V_{12} = V_3 = 10.53 V$

$$Q_3 = C_3 V_3 = (6 \mu)(10.53) = 63.2 \mu C$$

$$Q_{12} = C_{12} V_{12} = (2.5 \mu)(10.53) = 26.33 \mu C$$

C_1 and C_2 are in series so $Q_{12} = Q_1 = Q_2 = 26.33 \mu C$

$$\therefore Q_1 = 26.33 \mu C; Q_2 = 26.33 \mu C; Q_3 = 63.2 \mu C; Q_4 = 89.4 \mu C$$

- c) C_1 and C_2 are in series so $V_{12} = V_1 + V_2 = 10.53 V$

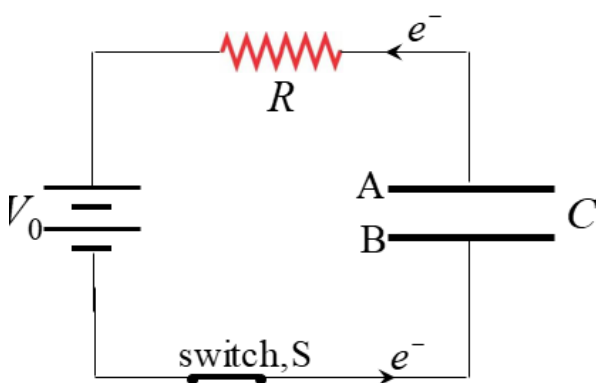
$$V_1 = \frac{Q_1}{C_1} = \frac{26.33 \mu}{15 \mu} = 1.76 V$$

$$V_2 = \frac{Q_2}{C_2} = \frac{26.33 \mu}{3 \mu} = 8.78 V$$

$$\therefore V_1 = 1.76 V; V_2 = 8.78 V; V_3 = 10.53 V; V_4 = 4.47 V$$

4.2 Charging And Discharging Of Capacitors

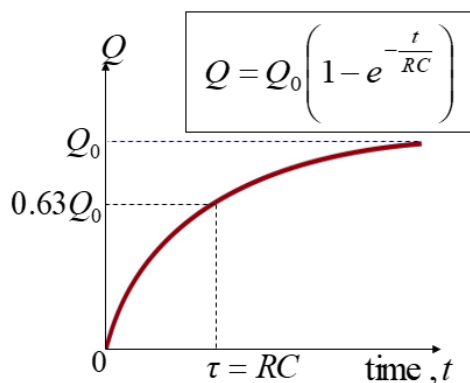
Charging of capacitor



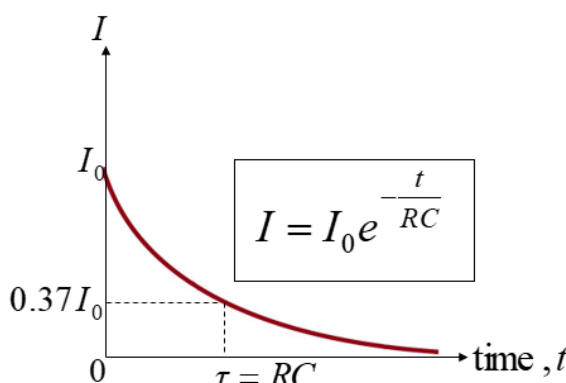
Simulation:



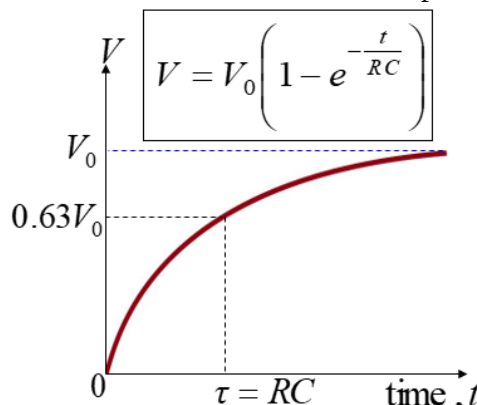
- When switch S is closed, current I_0 immediately begins to flow through the circuit.
- Electrons will flow out from the negative terminal of the battery and accumulate on the plate B of the capacitor.
- Then electrons will flow into the positive terminal of the battery through the resistor R, leaving a positive charge on the plate A.
- As charges accumulate on the capacitor, the potential difference across it increases and the current is reduced until eventually the maximum voltage across the capacitor equals the voltage supplied by the battery, V_0 .
- At this time, no further current flows ($I = 0$) through the resistor R and the charge Q on the capacitor thus increases gradually and reaches a maximum value Q_0 .



The charge on the capacitor increases with time.

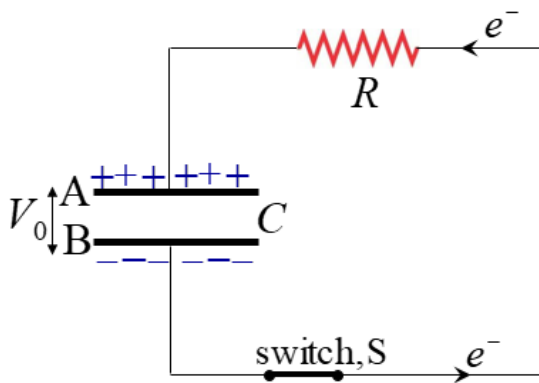


The current through the resistor decreases exponentially with time.



The potential difference across capacitor increases with time.

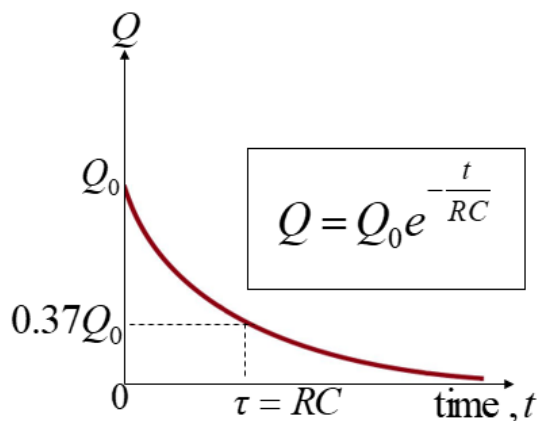
Discharging of capacitor



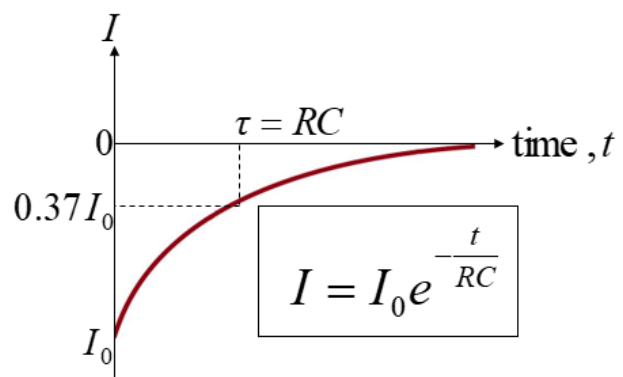
Simulation:



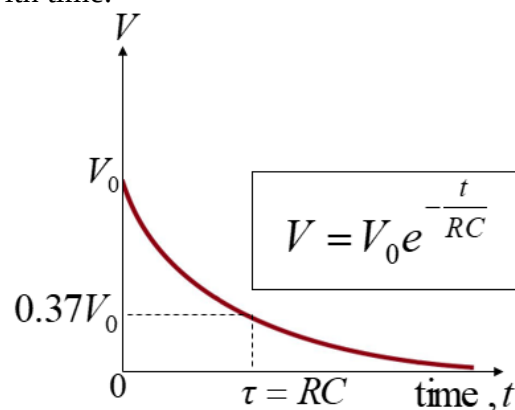
- When switch S is closed, electrons from plate B begin to flow through the resistor R and neutralizes positive charges at plate A.
- Initially, the potential difference (voltage) across the capacitor is maximum, V_0 and then a maximum current I_0 flows through the resistor R.
- When part of the positive charges on plate A is neutralized by the electrons, the voltage across the capacitor is reduced.
- The process continues until the current through the resistor is zero.
- At this moment, all the charges at plate A is fully neutralized and the voltage across the capacitor becomes zero.



The charge on the capacitor decreases exponentially with time.



The current through the resistor decreases exponentially with time.



The potential difference across capacitor decreases exponentially with time.

Time constant

The quantity RC that appears in the exponent for all equations in the charging and discharging process is called time constant, τ

$$\tau = RC$$

τ : time constant

R : Resistance

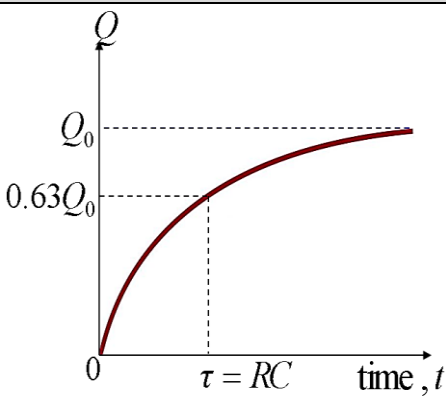
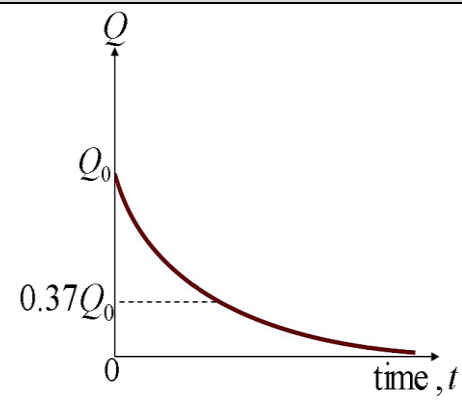
C : Capacitance

It is a scalar quantity and the S.I. unit is second (s).

Time constant a measure of how quickly the capacitor charges or discharges.

Do not use the equation to define time constant, τ .

Physical meaning of time constant

Charging	Discharging
 Time constant, τ is defined as the time required for the capacitor to reach 0.63 or 63% of its maximum charge.	 Time constant, τ is defined as the time required for the charge on the capacitor to decrease to 0.37 or 37% of its initial (maximum) value.

Example 5

A $2\mu\text{F}$ capacitor is charged to 10 V, and then discharged through a $1.0\text{ M}\Omega$ resistor. Calculate

- The time constant of the discharge circuit.
- The time taken for the potential difference across the capacitor to drop to 5 V.

Answers:

a) $R = 1.0 \times 10^6 \Omega$; $C = 2.0 \times 10^{-6} \text{ F}$

$$\tau = RC$$

$$\tau = (1.0 \times 10^6)(2.0 \times 10^{-6})$$

$$\tau = 2.0 \text{ s}$$

b) For discharging process,

$$V = V_0 e^{-\frac{t}{RC}}$$

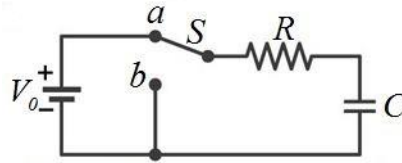
$$5 = (10)e^{-\frac{t}{2}}$$

$$0.5 = e^{-\frac{t}{2}}$$

$$t = 1.39 \text{ s}$$

Example 6

In the RC circuit shown in figure below, the battery has fully charged the capacitor.



Then at $t = 0$ s the switch S is thrown from position a to b . Given the voltage of the battery is 20.0 V, the resistance of the resistor is $57\ \Omega$ and the capacitance $C = 1.02\ \mu\text{F}$. Determine

- The time constant, τ .
- The value of the charge, Q on the capacitor at $t = 0$ s.
- The value of Q at $t = 60\ \mu\text{s}$.

Answers:

$$\begin{aligned} \text{a) } R &= 57\ \Omega; C = 1.02 \times 10^{-6}\text{ F} \\ \tau &= RC \\ \tau &= (57)(1.02 \times 10^{-6}) \\ \tau &= 5.81 \times 10^{-5}\text{ s} \end{aligned}$$

$$\begin{aligned} \text{b) } Q_0 &= CV_0 \\ Q_0 &= (1.0 \times 10^{-6})(20.0) \\ Q_0 &= 2.04 \times 10^{-5}\text{ C} \end{aligned}$$

- For discharging,

$$Q = Q_0 e^{-\frac{t}{RC}}$$

$$\begin{aligned} Q &= (2.04 \times 10^{-5})e^{-\frac{60 \times 10^{-6}}{5.81 \times 10^{-5}}} \\ Q &= (2.04 \times 10^{-5})e^{-1.05} \\ Q &= 7.26\ \mu\text{C} \end{aligned}$$

Chapter 4 MCQs Practice:**SELECTED FORMULAE**

- $C = \frac{Q}{V}$
- $\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$
- $C_{eff} = C_1 + C_2 + C_3 \dots C_n$
- $\tau = RC$
- $Q = Q_0 e^{-\frac{t}{RC}}$
- $Q = Q_0(1 - e^{-\frac{t}{RC}})$

DIY Quiz

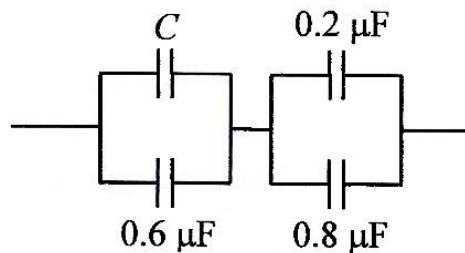
Instruction:

Answer all questions in 10 minutes.

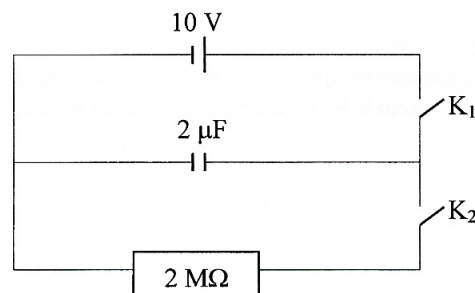
No discussion & do not refer to any notes.

You can only refer to the formulae given below.

1. Figure shows an arrangement of capacitors. If the equivalent capacitance is $0.5 \mu\text{F}$, calculate the value of capacitor C .



2. Figure shows a $2 \mu\text{F}$ capacitor connected to a 10 V battery, a $2 \text{ M}\Omega$ resistor and switches K_1 and K_2 .

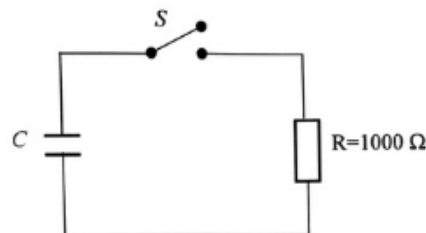


- a) Calculate the charge of the capacitor when switch K_1 is closed and switch K_2 is opened.
 b) Switch K_1 is then opened and switch K_2 closed. Calculate the charge in the capacitor, 8.0 s after K_2 is closed.

Answers:

1. $C = 4.0 \times 10^{-7} \text{ F}$

2. a) $Q = 20 \mu\text{C}$; b) $Q = 2.71 \mu\text{C}$

PAST YEAR QUESTIONS

A fully charged capacitor, C is connected to a resistor, $R = 1000 \Omega$ and a switch, S as shown in figure above. The time constant, τ for the circuit is 2.50 s . Switch S is then closed and the charge on the capacitor becomes 50% of its initial value at time, t .

- a) Calculate capacitance of the capacitor.
 b) Calculate value of t .
 c) Sketch graph of magnitude of current against time for this process.

[5 marks]